

Assignment 6: Triage Pseudocode

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Design goal

This is an instruction-based design for a first-pass digital triage screen. The system reads the cleaned Week 0 patient fields, checks whether the values make sense, applies threshold rules, and returns a risk level with the reasons. It does not diagnose the patient or replace the triage nurse. It helps the nurse see who may need attention first.

Inputs used

The Week 0 dataset gives enough fields for a basic risk screen: ID, Age, GCS, SBP, DBP, MAP, pulse, Temp, RR, and FiO2. I would add SpO2 later, because oxygen saturation tells us whether the patient is getting enough oxygen in the blood. FiO2 shows the oxygen being delivered, not the oxygen in the blood.

Risk levels

- High risk: immediate clinical review. One severe vital sign can be enough.
- Moderate risk: repeat vitals, review symptoms, and reassess soon.
- Low risk: no clear abnormal threshold from the available vitals.
- Needs manual review: the data is missing, impossible, or internally inconsistent.

Pseudocode

```
FOR each patient record:
```

```
  Start risk_level as "Low"
```

```
  Start reasons as an empty list
```

```
  Start data_issues as an empty list
```

```
  Read ID, Age, GCS, SBP, DBP, MAP, pulse, Temp, RR, FiO2
```

```
  Check data quality first:
```

```
    If a required value is missing, add it to data_issues
```

```
    If GCS is outside 3 to 15, add "invalid GCS"
```

```
    If SBP or DBP is not possible, add "invalid blood pressure"
```

```
    If DBP is greater than SBP, add "blood pressure values reversed or incorrect"
```

```
    If MAP does not match SBP and DBP closely, recalculate MAP or flag it
```

```
    If pulse, Temp, or RR is outside a possible human range, flag it
```

```
  If data_issues contains a value needed for scoring:
```

```
    Set risk_level to "Needs manual review"
```

```
    Output ID, risk_level, and data_issues
```

```
    Move to the next patient
```

```
  Check high-risk rules:
```

```
    If GCS is 8 or lower, add "severely reduced consciousness"
```

```
    If SBP is 90 or lower, or MAP is below 65, add "possible poor perfusion"
```

```
    If RR is 8 or lower, or RR is 30 or higher, add "dangerously abnormal breathing rate"
```

```
    If pulse is below 40 or 130 or higher, add "dangerously abnormal heart rate"
```

```
    If Temp is below 35.0 C or 40.0 C or higher, add "dangerous temperature"
```

```
    If SpO2 is available and below 90%, add "low oxygen saturation"
```

```
  If reasons has at least one high-risk reason:
```

```
    Set risk_level to "High"
```

```
    Output ID, risk_level, reasons, and "clinical review now"
```

```
    Move to the next patient
```

```
  Check moderate-risk rules:
```

```
    Start moderate_count at 0
```

```
    If GCS is 9 to 12, add one to moderate_count
```

```

    If SBP is 91 to 100, add one to moderate_count
    If pulse is 101 to 129, or 40 to 50, add one to moderate_count
    If RR is 21 to 29, or 9 to 11, add one to moderate_count
    If Temp is 38.1 C to 39.9 C, or 35.0 C to 36.0 C, add one to moderate_count
    If FiO2 is 40% or higher, add one to moderate_count
    If Age is 65 or older and another abnormal sign exists, add one to moderate_count

    If moderate_count is 2 or more:
        Set risk_level to "High"
        Add "multiple abnormal signs together" to reasons
    Else if moderate_count is 1:
        Set risk_level to "Moderate"
        Add "one abnormal sign needs reassessment" to reasons
    Else:
        Keep risk_level as "Low"
        Add "no threshold crossed in available fields" to reasons

    Output ID, risk_level, reasons, and next action

```

Threshold justification

I used simple thresholds because the Week 0 dataset does not include final diagnosis, ED outcome, or clinician-assigned triage category. A rule-based design is easier to explain and safer for this exercise than training a predictive model without outcome labels. The system also returns the reason for each flag, so a nurse can check the decision instead of seeing a label with no explanation.

The normal adult ranges came from basic vital sign references: pulse around 60 to 100 bpm, respiratory rate around 12 to 18 breaths per minute, blood pressure above 90/60 and below 120/80, and temperature around 36.5 C to 37.3 C. I used more urgent cutoffs when the value suggests possible shock, respiratory distress, altered consciousness, or severe fever. NEWS2-style early warning logic also supports using extreme values in respiratory rate, systolic blood pressure, pulse, consciousness, oxygen status, and temperature to trigger escalation.

Why the flow works

The system checks data quality before scoring because bad data should not become a low-risk result. It then checks severe red flags before moderate ones, because a single dangerous value can matter more than an average-looking set of vitals. Age does not create high risk by itself. Age raises concern when another abnormal sign is present, since older patients may have less reserve when they deteriorate.

AI tool use

I used AI tools to pressure-test the flow, compare the draft against the rubric, and check whether the pseudocode was easy to follow. I did not use AI output as clinical authority. I checked the threshold ideas against medical references and kept the final design rule-based, visible, and limited to the Week 0 dataset.

References

MedlinePlus, Vital Signs: <https://medlineplus.gov/ency/article/002341.htm>

Royal College of Physicians, National Early Warning Score (NEWS2): https://www.rcp.ac.uk/media/a4ibkbbf/news2-final-report_0_0.pdf

NCBI Bookshelf, Glasgow Coma Scale: <https://www.ncbi.nlm.nih.gov/books/NBK513298/>

NCBI Bookshelf, Hypotension: <https://www.ncbi.nlm.nih.gov/books/NBK499961/>

MedlinePlus, Pulse Oximetry: <https://medlineplus.gov/lab-tests/pulse-oximetry/>